



A Capitalized Title: Something about a Topic

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Abstract

This guide of the Austrian Journal of Statistics (AJS) shows the required style of manuscripts submitted to the Austrian Journal of Statistics. We strongly recommend to follow the given instructions exactly already when writing the paper. When submitting papers to AJS, it is essential to follow these guidelines. Manuscripts submitted without consideration of the AJS style guide will most likely not be forwarded for review, but rejected with the possibility of resubmission.

Keywords: keywords, not capitalized, Java.

1. Placeholder!! can be deleted

2. Data and Methods

2.1. Data Description

This project is based on genetic and osteological data obtained from human remains recovered from Mycenaean collective burial contexts in Greece, dated to approximately 1500–1000 BCE. The material derives from two archaeological sites: Amfissa, which consists of a single tholos tomb, and Elateia, a large chamber tomb complex containing 85 chambers. From Elateia, eight chambers were selected for detailed genetic investigation. DNA preserved in skeletal material was extracted and analyzed to infer biological relationships among individuals buried within and between tombs.

The analysis draws on three interconnected datasets that capture information at different levels of resolution. One dataset summarizes characteristics at the tomb level. It includes variables such as the minimum number of individuals identified, the number of individuals successfully analyzed genetically, the estimated duration over which the tomb was used, the proportion of successful DNA samples, the total number of samples taken, and the distribution

of sampled skeletal elements (including petrous bones, teeth, talus or phalanx, and other elements). These variables highlight differences between tombs in preservation, sampling, and burial practices.

A second dataset contains metadata describing the sampled individuals. Each entry represents a single individual and records the associated tomb, estimated biological sex, age category, and the skeletal element used for DNA extraction. This information provides the biological and contextual background necessary to evaluate how individual characteristics relate to genetic data quality and inferred relationships.

The third dataset reports pairwise genetic relationships between individuals based on ancient DNA analyses. For each pair, it includes estimates of genetic relatedness, the number of overlapping single nucleotide polymorphisms (SNPs), and a categorical classification of relationship degree following established criteria. Since individuals can be involved in multiple pairwise comparisons, observations in this dataset are not independent, resulting in a complex relational structure.

2.2. Data Preparation for Model

Prior to statistical modeling, the raw datasets were processed and integrated to obtain consistent pairwise datasets suitable for analysis. Data preparation involved filtering relevant observations, harmonizing variable definitions, combining information across different levels, and constructing derived variables required for modeling.

At the individual level, metadata were first restricted to the burial sites and tombs included in the analysis. Individuals from Lapoutsi were excluded, and only samples from Amfissa tholos and selected tombs at Elateia were retained. Age estimates were harmonized by collapsing ambiguous or overlapping categories into two broad groups, namely subadult and adult or undefined, to ensure sufficient sample sizes per category.

Tomb-level parameters were prepared separately and standardized to ensure consistency across datasets. These parameters include the minimum number of individuals, the number of genetically analyzed individuals, the estimated duration of tomb use, and the proportion of successful DNA samples. To allow analyses at different spatial resolutions, an additional set of tomb-level parameters was constructed in which Tombs 46, 56, and 62 at Elateia were treated as a single combined burial unit. Corresponding values for this grouped tomb were derived from the original data and added alongside the individual tomb records.

Pairwise genetic kinship data were then processed to generate the main analytical datasets. Relationship classifications were cleaned by distinguishing between confidently assigned relationships and uncertain cases. Pairs with insufficient genetic information were marked as uncertain, and a binary relatedness indicator was derived that distinguishes related and unrelated pairs while excluding uncertain observations. In addition, a global relationship classification based on a stricter minimum SNP overlap threshold of 15,000 was created to reduce bias associated with low-quality genetic data.

Individual-level metadata were merged into the kinship dataset for both members of each pair, adding information on tomb affiliation, sex, age category, and sampled skeletal element. Based on these merged data, two parallel analytical datasets were constructed. The first dataset retains the original tomb structure, in which Tombs 46, 56, and 62 are treated as separate burial contexts. The second dataset includes an additional set of observations in which pairs involving individuals from these tombs are duplicated and reassigned to a combined tomb category. This approach preserves information at the individual tomb level while enabling sensitivity analyses that assess the impact of treating closely related burial contexts as a single unit.

Finally, both datasets were augmented with tomb-level characteristics by linking each individual's tomb to the corresponding sampling success, number of analyzed individuals, and length of use. From these variables, pairwise summaries were derived, including the minimum

sampling success, the minimum number of analyzed individuals, and the absolute difference in tomb use duration between the two burial contexts. Additional pair-level variables were constructed to describe sex composition, age composition, shared tomb membership, combined skeletal element types, and an indicator of bone quality based on the presence of petrous bone samples.

The resulting datasets consist of pairwise observations enriched with individual- and tomb-level information and derived covariates. Together, they allow the evaluation of genetic relatedness patterns under alternative representations of burial organization and provide the basis for the multi-membership modeling strategy applied in the subsequent analysis.

2.3. The Multi-Membership Generalized Linear Mixed Model

Model Definition and Motivation

To analyze the kinship patterns within the dataset, we require a modeling framework that can account for the inherent dependencies within our observations. In this study, an observation is defined as a pair of individuals (i, j) , where the response variable Y_{ij} indicates whether the pair shares a biological relationship.

Motivation: The Complexity of Pairwise Data

The data structure presents two primary challenges that traditional regression models cannot handle: Multiple Membership of Individuals: Each individual appears in multiple pairs. For example, if individual i is compared to j , k , and l , the errors associated with those three observations will be correlated because they all share the unique genetic and environmental traits of individual i . Tomb-Level Dependencies: Pairs can exist both within the same tomb or across different tombs. Because individuals within a tomb may share common environmental or social factors, we must account for the random “tomb effect.” A pair i, j effectively “belongs” to the tombs $T(i)$ and $T(j)$.

To address this, we utilize a Multi-Membership Generalized Linear Mixed Model (MM-GLMM). This model is specifically designed for cases where an observation does not belong to a single cluster, but rather “members” multiple clusters (in our case, two individuals and two tombs) simultaneously [Browne, Goldstein, and Rasbash \(2001\)](#).

Model Definition

Let Y_{ij} be a binary indicator where $Y_{ij} = 1$ if individuals i and j are related, and $Y_{t,ij} = 0$ otherwise. We assume Y_{ij} follows a Bernoulli distribution:

$$Y_{t,ij} \sim \text{Bernoulli}(\pi_{ij})$$

where $\pi_{ij} = P(\text{related}_{ij} = 1)$ is the probability of a relationship existing between the pair. We model this probability using a logit link function to accommodate the complex random effects structure:

$$\text{logit}(\pi_{ij}) = \alpha + \mathbf{X}_{ij}^\top \boldsymbol{\beta} + \underbrace{(w_i u_i + w_j u_j)}_{\text{Individual RE}} + \underbrace{(w_{T(i)} v_{T(i)} + w_{T(j)} v_{T(j)})}_{\text{Tomb RE}}$$

Components of the Model:

- α : The global intercept, representing the baseline log-odds of two individuals being related.
- $\mathbf{X}_{ij}\boldsymbol{\beta}$: The fixed effects component, where $\mathbf{X}_{ij}$ contains pair-level covariates.
- u_i, u_j : Random effects for the two individuals, where $u \sim \mathcal{N}(0, \sigma_{\text{ind}}^2)$
- $v_{T(i)}, v_{T(j)}$: Random effects for the tombs associated with individuals i and j , where $v \sim \mathcal{N}(0, \sigma_{\text{tomb}}^2)$

- w : Weights assigned to the random effects. Following standard multi-membership notation [Browne et al. \(2001\)](#), we set equal weights $w_i = w_j = 0.5$ and $w_{T(i)} = w_{T(j)} = 0.5$. This ensures that the total variance contributed by the individual or tomb level remains consistent and equal.

Fitting the Model with brms

Following the theoretical definition of the MM-GLMM, this section details the practical implementation, predictor selection, and the Bayesian framework used for estimation.

Predictor Selection and the Causal Framework The selection of covariates is guided by the Directed Acyclic Graph (DAG) presented in Section [2.3.1. DAG]. To estimate the probability of relatedness, we focus on variables situated along the social, biological branches of our causal model. Specifically, the following predictors are included in the model: Pairwise Sex Combination, Pairwise Age Category, and Same Tomb Indicator.

Notably, DNA preservation variables (Number of Overlapping SNPs, Analyzed skeletal element,...) are excluded from the model. While preservation significantly influences data quality and the initial ability to identify SNPs, it does not exert a causal influence on true biological relatedness itself. Because a rigorous quality filter, a 15,000 SNP threshold, was applied during data preprocessing, the influence of preservation bias is effectively mitigated. Once a pair meets this threshold, variation in kinship outcomes is no longer explained by preservation quality.

Bayesian Implementation via brms The model is fitted using the *brms* package in R. The *brms* package acts as an interface to Stan, utilizing Hamiltonian Monte Carlo (HMC) algorithms to perform Bayesian inference. This approach is particularly robust for Multi-Membership models, as it allows for flexible specification of complex random effect structures that often struggle to estimate using frequentist Maximum Likelihood methods [Bürkner \(2017\)](#).

Two distinct versions of the model are implemented

Separated Tomb Model: Every tomb is treated as an individual level within the random effect v_t . Grouped Tomb Model: In this version, three specific tombs, Elateia T46, T56, and T62, are treated as a single pooled tomb.

3. Result

Following the model specification, this chapter presents the findings from the Bayesian Multi-Membership Generalized Linear Mixed Model.

3.1. Model Diagnostics

Before interpreting the relationship between burial patterns and kinship, it is essential to assess the “sampling health” of the model. Because the MM-GLMM was fitted using Hamiltonian Monte Carlo (HMC) via Stan, we evaluate the reliability of the posterior distributions using two primary metrics: the potential scale reduction factor ($\hat{R}$) and the Effective Sample Size (ESS).

Convergence and Chain Stability: A primary indicator of model health is the $\hat{R}$ (R-hat) statistic, which compares the variance within individual MCMC chains to the variance between chains. For all parameters in our model, the $\hat{R}$ values were consistently **1.00**. Following modern Bayesian standards, an $\hat{R} < 1.01$ indicates that all four chains have successfully converged to the same posterior distribution, confirming that the model has reached stability and is not sensitive to the starting values of the chains [Vehtari, Gelman, Simpson, Carpenter, and Bürkner \(2021\)](#).

Sampling Efficiency: The Effective Sample Size (ESS) provides an estimate of the number of independent draws contained in the posterior distribution, accounting for autocorrelation

between consecutive MCMC steps. In accordance with the standards implemented in the brms framework, we report two measures:

- Bulk-ESS: Measures the sampling efficiency for the mean and median of the distribution.
- Tail-ESS: Measures the efficiency for the tails of the distribution, which is critical for the reliability of the 95% credible intervals.

As shown in the model output, both the Bulk-ESS and Tail-ESS for all key predictors are well above the recommended thresholds (typically >400 per chain). Many parameters reached between 3,000 and 22,000 independent samples. For instance, the “Same Tomb” predictor achieved a Bulk-ESS of 22,386, indicating an exceptionally high degree of precision in the estimation of the interment effect. These metrics provide a high degree of confidence that the posterior estimates discussed in the following sections are robust and represent truly independent information from the data [Vehtari *et al.* \(2021\)](#).

3.2. Model Output

Following the diagnostic verification, we present the posterior estimates for the two versions of our model: the Separated Tomb Model (8 tombs) and the Grouped Tomb Model (6 tombs). Both versions show remarkable stability in their regression coefficients, confirming that the pooling of the Elateia tombs does not fundamentally alter the estimated relationships. For a visual representation of the full effect estimations and their associated 95% Credible Intervals (CI), please refer to the plot in ??Figure.

A primary observation is the dominance of Same Tomb Indicator, which yielded the highest estimate in both models (2.30 and 2.64, respectively). This signifies that shared burial space is the strongest predictor of biological kinship in this dataset. Regarding biological predictors, the Male-Male (XY-XY) combination stands out with a consistent estimate of 1.94, significantly higher than the Female-Male (XX-XY) estimate of ≈ 0.90 . Furthermore, the Both Subadult category (1.51 and 1.46) suggests a strong horizontal kinship signal among children. The multilevel hyperparameters reveal substantial variation at the individual level ($sd \approx 2.29$), which suggests that the density of kinship ties is not uniform across the population, but rather concentrated around specific individuals. At the tomb level, the estimated standard deviation increases from 1.16 in the Separated Model to 1.86 in the Grouped Model. This increase suggests that grouping the Elateia (T46, T56, and T62) tombs allows the model to better account for differences between burial sites. Crucially, the fixed effects remained consistent across both models, proving that the results are not dependent on how the tombs were categorized.

3.3. Model Interpretation

The model output is reported in log-odds, as is standard in logistic regression. However, coefficients expressed on the log-odds scale are not directly intuitive to interpret. To make the results more accessible, we therefore calculate marginal predicted probabilities. These convert the model estimates into probabilities of the outcome, which in this case, the probability that a pair of individuals is biologically related.

Marginal Effect (Predicted Probabilities):

Marginal predicted probabilities indicate how each explanatory variable affects the average probability of relatedness, while holding all other variables constant (*ceteris paribus*). This approach allows for a more accessible interpretation of the model results by expressing effects in terms of probabilities rather than log-odds.

??? me plots

The average predicted probabilities in the figure highlight several key patterns across sex pairings, age compositions, and burial contexts. Female–female pairs show relatively low average predicted probabilities of relatedness (0.21% in the separate tombs model and 0.31% in

the grouped tombs model, *ceteris paribus*), while male–male pairs exhibit higher probabilities compared to other sex combinations. For age composition, pairs consisting of two subadults display higher predicted probabilities than mixed or adult–adult pairs. The most pronounced effect, however, is observed for burial context: pairs interred in the same tomb show substantially higher predicted probabilities of relatedness than any other category in both model specifications

3.4. Model Robustness Check

To ensure that the results obtained in the primary MM-GLMM are not artifacts of specific data distributions, prior specifications, or classification choices, we performed three sensitivity analyses. These tests evaluate whether our findings, particularly the dominant role of the tomb effect and sex-specific patterns, remain consistent under alternative model conditions.

Sensitivity to Prior Specification

In the primary model, we utilized flat (uninformative) priors, which allow the data to drive the estimates entirely. To ensure the model was not “noise chasing”, which means fitting patterns that do not truly exist, we re-fitted the model using Weakly Informative Priors $N(0, 1)$. This approach applies a form of regularization, pulling estimates toward zero unless the data provides strong evidence to the contrary.

As expected with Priors $N(0, 1)$, the coefficients experienced a slight shrinkage toward zero. No coefficients changed direction, and the variable Same Tomb Indicator remained the dominant predictor, largely unaffected by the prior. This confirms that the observed effects are driven by strong signals in the data rather than being an artifact of the prior choice.

Sensitivity to the “Same Tomb” Predictor (Within-Tomb Analysis)

Because the Same Tomb Indicator emerged as an exceptionally dominant predictor in our primary model, it was necessary to conduct a robustness check to ensure that other biological and demographic variables were not merely statistical artifacts of this effect.

To isolate the influence of sex and age from the overwhelming signal of shared burial space, we fitted a secondary model restricted exclusively to within-tomb pairs. This approach effectively removes the across-tomb variance, providing a focused look at the internal organization of the burial units. The model was restricted to the 2,990 observations involving individuals interred in the same tomb, compared to the 13,690 observations in the full dataset. Consequently, the estimated effects appear weaker, with coefficients shrinking toward zero.

Despite the reduced power, the qualitative results remained consistent. The direction of all effects was unchanged, and the Male-Male (XY-XY) pair combination remained the dominant predictor of relatedness within the tomb. This confirms that the sex-biased patterns identified in the primary analysis are a fundamental characteristic of the kinship structure within these tombs and are not driven by the broader “Same Tomb” effect.

Redefining the Relatedness Threshold (3rd Degree as Unrelated)

In our primary analysis, we collapsed 1st, 2nd, and 3rd-degree relatives into a single “Related” category. However, 3rd-degree relationships represent more distant biological ties. We conducted a robustness check by redefining the response variable Y_{ij} to count only 1st and 2nd-degree relatives as “Related.”

Under this stricter definition, the “Same Tomb” effect showed a substantially larger estimated coefficient. This suggests that shared burial context is an even stronger predictor for immediate kinship than for distant relatives. From an archaeological perspective, 3rd-degree kin may not have resided in the same immediate household or been interred in the family crypt as strictly as parent-child pairs. By including 3rd-degree relatives in the primary “Related” category, we effectively “diluted” the interment effect.

4. Discussion

4.1. Model assumption

While the MM-GLMM provides a robust framework for analyzing pairwise kinship, the validity of its inferences depends on the underlying statistical assumptions. This section discusses the theoretical and empirical justifications for our model choices, as well as the potential impacts of deviations from these assumptions.

Equal Weighting of Random Effects ($w = 0.5$)

In the multi-membership structure, we assumed equal weights ($w_i = w_j = 0.5$ and $w_{T(i)} = w_{T(j)} = 0.5$) for the random effects of individuals and tombs, respectively. This assumption is theoretically plausible at the individual level, as there is no a priori reason to believe one individual in a pair contributes more to the shared probability of relatedness than the other. Furthermore, simulation studies by [Smith and Beretvas \(2014\)](#) suggest that multi-membership models are generally robust to weight misspecification, meaning that even if the “true” weights were slightly unequal, the resulting fixed-effect estimates remain reliable.

Normality of Random Effects

A standard assumption of GLMMs is that the random effects (u and v) are independently and identically distributed following a normal distribution: $u, v \sim \mathcal{N}(0, \sigma^2)$. We evaluated this through Q-Q plots for both the Individual and Tomb random effects across our model variants.

For individual random effect diagnostic plots, the distribution is somewhat skewed. Non-normality in random effects can potentially lead to a biased global intercept and an inaccurate estimation of the random effect distribution’s shape. However, literature on mixed models [McCulloch and Neuhaus \(2011\)](#) demonstrates that fixed-effect estimates and variance components are stable even when the normality assumption for random effects is violated.

The diagnostic for the tomb-level effects is limited by the low number of levels. According to [Maas and Hox \(2005\)](#), the estimation of the random effect variance (σ_{tomb}^2) can be biased or imprecise when the number of clusters is small (typically < 30). However the fixed-effect regression coefficients remain unbiased. The model therefore remains a valid tool to interpret the influence of the fixed-effect variables.

??? Put QQ Plots

4.2. Selection Bias

A threshold of 15,000 overlapping SNPs was applied across all kinship degrees to address the issue of missing not at random (MNAR) data. By restricting the analysis to observations that meet this quality criterion, the focus is placed exclusively on a high-quality subset of the available genetic data. Although this procedure increases the reliability of kinship estimates, it may introduce selection bias, as individuals with lower data quality are systematically excluded from the analysis. As a result, the findings may not be fully generalizable to the broader population under study.

4.3. Outlook for Future Studies

Future research should aim to expand the empirical basis of the analysis by incorporating a substantially larger number of tombs across multiple archaeological contexts. A broader sample would allow for the estimation of robust tomb-level or multilevel models with a sufficient number of higher-level units. This would enable a more reliable assessment of tomb-specific characteristics—such as length of use, number of interred individuals, and skeletal preserva-

tion patterns—and their influence on genetic relatedness structures. With increased statistical power, contextual variation between burial units could be modeled more systematically.

In addition, future research should further address the issue of missing data. In the present study, applying a single SNP threshold helped reduce bias in the classification of kinship degrees. However, this decision also meant that only individuals with sufficiently high genetic data quality were included in the analysis. As a result, part of the available information had to be excluded. Future studies could benefit from improved sequencing quality and broader genomic coverage, which would reduce the need for strict thresholds and allow more individuals to be retained in the analysis. Moreover, applying alternative statistical approaches to evaluate how different missing data assumptions affect the results could help assess the stability and robustness of the findings.

From a methodological perspective, future research could explore extensions of the multi-membership generalized linear mixed model. Relaxing assumptions such as equal random-effect weights or normally distributed random effects may allow for a more flexible representation of individual- and tomb-level heterogeneity. Furthermore, larger datasets would permit more stable estimation of variance components, particularly at the tomb level, thereby improving inference regarding contextual effects.

Finally, future work should evaluate the potential impact of selection processes introduced by quality thresholds. Comparing results obtained from high-quality subsamples with those from broader inclusion criteria or alternative imputation strategies may help assess the generalizability of the findings to the wider population under study.

5. A Acknowledgment

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6. B Use of AI tools

###Report ChatGPT (GPT-5.2) were used to support the drafting and revision of the written report. The tools assisted in improving clarity, structure, and linguistic precision, particularly in reformulating paragraphs, refining academic wording, and ensuring coherence between sections. All conceptual content, interpretations, statistical reasoning, and conclusions were developed independently by the authors. The AI tool did not generate original scientific arguments but served as a language and structure support tool. All outputs were critically reviewed, edited, and verified by the authors. ??Latex??

###Code ChatGPT (GPT-5.2) and Gemini? were used to assist in troubleshooting, refining, and optimizing statistical code (e.g., for logistic regression models and data visualization). This included suggestions for improving code structure, debugging errors, and clarifying model implementation. However, all analytical decisions, model specifications, variable selections, and interpretations of results were made independently. The correctness and functionality of all code were manually tested and validated before inclusion in the analysis.

###Literature Research ChatGPT (GPT-5.2) was used to support the development of literature search strategies and to clarify theoretical concepts. All academic sources cited in the report were independently identified, accessed through scholarly databases, and critically

evaluated. No references were included without independent verification. The interpretation and integration of the literature were conducted manually.

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